AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# M348
# Applied Statistical Modelling

## Midterm Examination (MTA) - Makeup
## Spring Semester 2024/2025

## Date: 22/April/2025

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|---|---|
| **Number of Exam Pages:** 4 (including this cover sheet) | **Time Allowed:** 2 Hours |

**Instructions:**

- Total Marks: 30
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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## PART 1: True or False Questions [5x 1=5 marks]

**Which of the following statements are true/false?**

a) In multiple regression, the random error terms are assumed to be independent and normally distributed with constant variance.
b) A smaller training MSE and a larger test MSE typically indicate that the model is underfitting, as it performs poorly on both the training and test data.
c) The alternative hypothesis in multiple regression tests whether at least one of the coefficients βj differs from zero, not whether all coefficients are equal to zero.
d) A higher degree polynomial regression is an example of a more flexible model in statistical learning, as it allows for more complex, nonlinear relationships
e) Stepwise regression involves fitting a model by adding or removing predictors based on their statistical significance.

## PART 2: Regression with a factor for the Insurance dataset [11 marks]

**ALL QUESTIONS ARE REQUIRED**

The insurance dataset contains information about individuals seeking medical insurance, with variables that capture personal details and their insurance charges. Each row represents a unique individual, including their age, sex, BMI (Body Mass Index), number of children, smoking status, geographical region, and the total insurance charges they incur. The dataset aims to explore how these variables might relate to the cost of insurance, providing insights into how demographics, lifestyle choices, and regional factors contribute to medical expenses.
In this part, we will fit a regression model for the response variable charges from the insurance dataset, using smoker as a factor. These variables are described as:

- Charges: This variable represents the total medical charges incurred by each individual.
- Smoker: The smoker variable is categorical, with values "yes" and "no", indicating whether the individual is a smoker or not. Smoker status is an important factor influencing insurance charges, with smokers typically facing higher insurance premiums due to increased health risks.

```
Residuals:
   Min     1Q Median     3Q    Max
-19221  -5042   -919   3705  31720

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   8434.3      229.0   36.83   <2e-16 ***
smokeryes    23616.0      506.1   46.66   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7470 on 1336 degrees of freedom
Multiple R-squared:  0.6198,    Adjusted R-squared:  0.6195
F-statistic:  2178 on 1 and 1336 DF,  p-value: < 2.2e-16
```



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1) Complete the code below to fit a regression model for the response variable charges using the factor smoker as an explanatory variable, storing the results in an object called modelFactorFit. Obtain a summary of the results from fitting the model. (2 pts)
    i. modelFactorFit <- lm()
    ii. summary()
2) Based on the summary output, interpret the estimated value of the intercept. What does it represent in the context of insurance charges? (1 pt)
3) Interpret the estimated coefficient for smokeryes. What does it imply about the expected difference in insurance charges between smokers and non-smokers? (1 pt)
4) Compute the predicted insurance charges for: (2 pts)
    i. A non-smoker
    ii. A smoker
5) State the null and alternative hypotheses for testing the significance of the smoking variable in the model. (1 pt)
6) Based on the p-value for smokeryes, what conclusion can be drawn about its effect on insurance charges? (1 pt)
7) If additional variables (e.g., age or BMI) were included in the model, how might the Adjusted R-Squared (0.6195) change? Explain. (1 pt)
8) List two assumptions that linear regression makes about the residuals. (2 pts)

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PART 3: Multiple regression for the Insurance dataset [14 marks]

ALL QUESTIONS ARE REQUIRED

In this part, we will use multiple regression to model the response variable, charges, from the insurance dataset with three explanatory variables: age, bmi, and children. The goal is to examine how each of these variables contributes to the prediction of insurance charges. These variables are described as:

- Age: Age represents the individual's age in years. It is often a significant factor in determining insurance charges, as older individuals typically face higher medical expenses and therefore higher insurance premiums.
- BMI (Body Mass Index): BMI is a numerical value derived from an individual's height and weight. It is used as an indicator of body fat and can be associated with health risks, where higher BMI values generally correlate with higher insurance charges due to increased health risks.
- Children: This variable indicates the number of children or dependents covered by the insurance. Having more children might lead to higher charges due to the increased healthcare costs associated with additional dependents.

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```
Residuals:
    Min      1Q  Median      3Q     Max
-13884   -6994   -5092    7125   48627

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -6916.24    1757.48  -3.935 8.74e-05 ***
age           239.99      22.29  10.767  < 2e-16 ***
bmi           332.08      51.31   6.472 1.35e-10 ***
children      542.86     258.24   2.102   0.0357 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11370 on 1334 degrees of freedom
Multiple R-squared:  0.1201,    Adjusted R-squared:  0.1181
F-statistic: 60.69 on 3 and 1334 DF,  p-value: < 2.2e-16
```

1) Complete the code below to fit the multiple linear regression model with response variable charges and explanatory variables age, bmi and children, and then display a summary of the results from fitting the model. (2 pts)

```
modelFit =lm( __~__)
summary(__)
```

2) What is the interpretation of the estimated coefficient for age? How does age affect the insurance charges based on the model? (1.5 pts)
3) What does the coefficient for bmi suggest about its relationship with insurance charges? (1.5 pts)
4) Based on the p-values of the coefficients, which variables have a statistically significant effect on insurance charges at the 0.05 significance level? (2 pts)
5) What does the Multiple R-squared value of 0.1201 tell us about the model's ability to explain the variation in insurance charges? (1 pt)
6) Given the p-value of the F-statistic is < 2.2e-16, what conclusion can you draw about the model as a whole? (1 pt)
7) Using the regression model, what would be the predicted insurance charges for someone who is 30 years old, has a BMI of 28, and has 2 children? (2 pts)
8) What does the Residual Standard Error of 11370 indicate about the model's accuracy in predicting insurance charges? (1 pt)
9) If there were multicollinearity issues, list one method you would use to address them. (1 pt)
10) If the residuals are found to be non-normally distributed, list one solution you could apply to address this issue. (1 pts)

End of questions

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